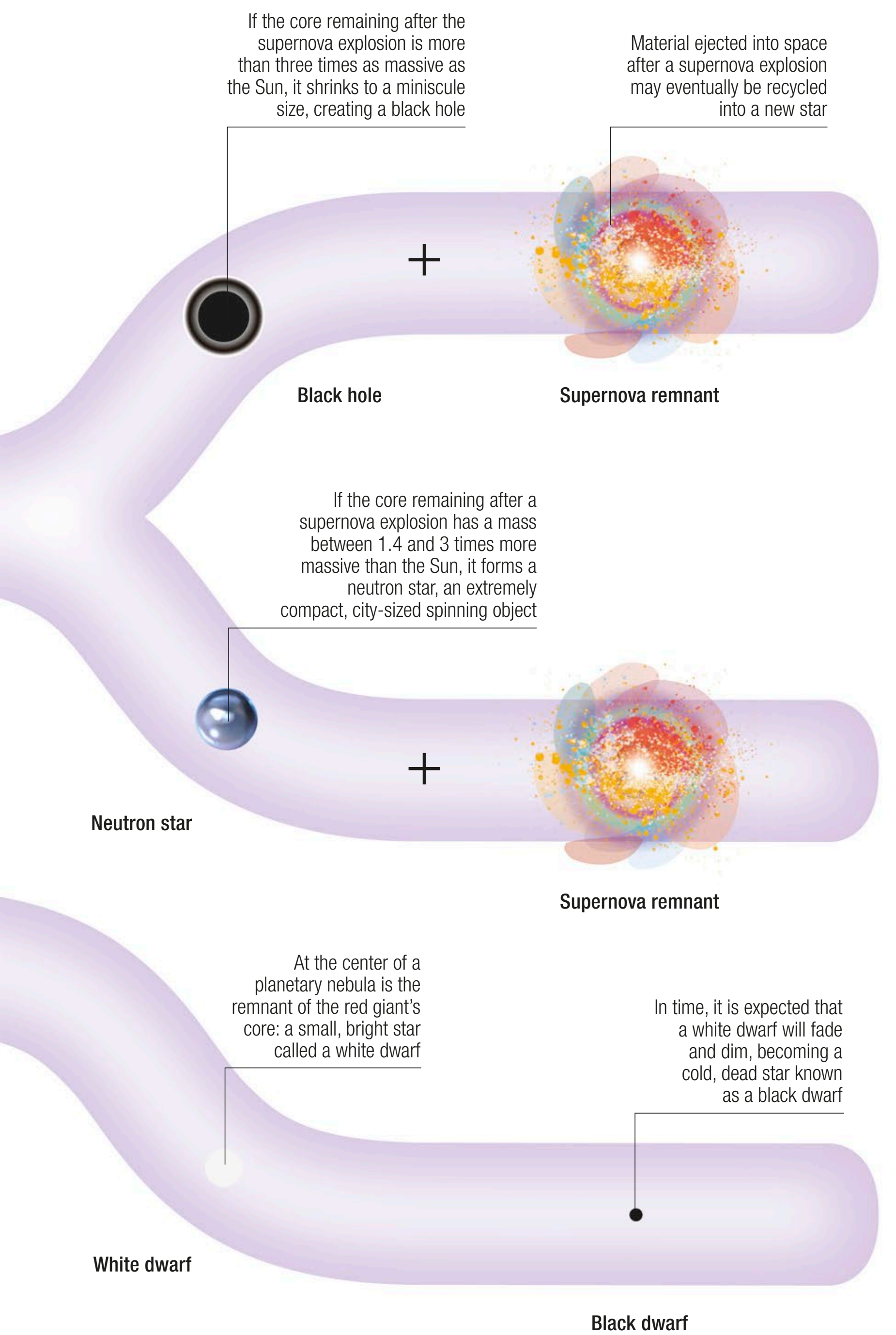
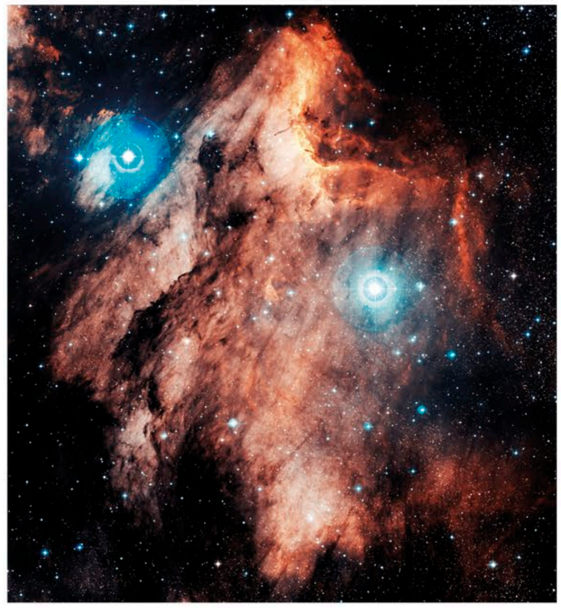
The **smallest red dwarf** stars can live **millions of times** longer than the largest hypergiant stars



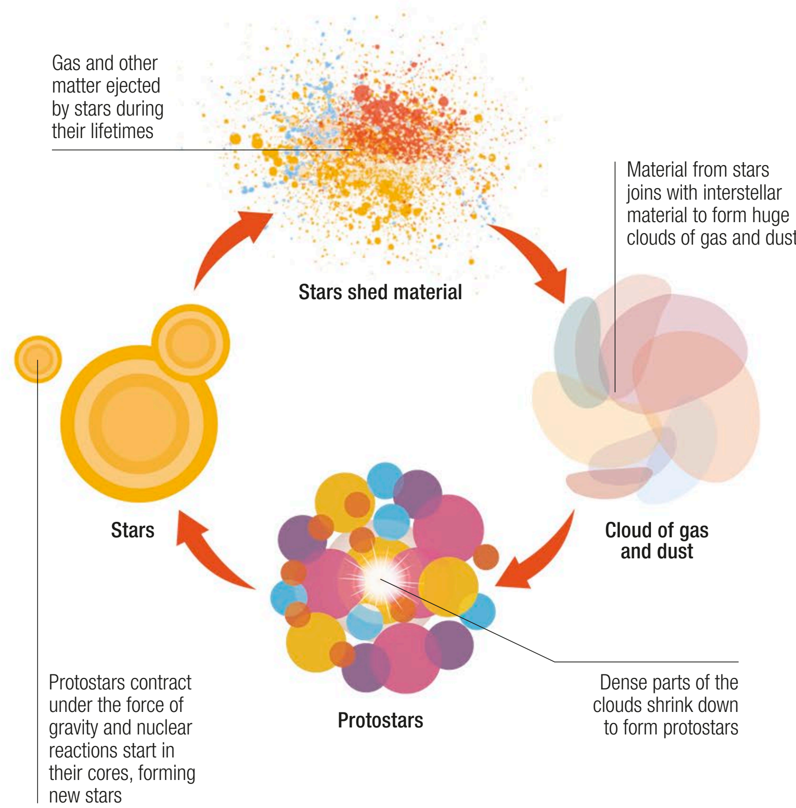
◁ **The lives of stars**
Contrasted here are the life stories of three main categories of stars: (from top) high-mass stars, medium-mass (Sun-sized) stars, and low-mass stars. Stars in each category start off as protostars that have formed in star-forming nebulae, but the course of their lives thereafter can be very different.



◁ **Star-forming region**
This site of intense star formation is known as the Pelican Nebula because part of it (near the top in this image) resembles the head of a pelican. It lies about 2,000 light-years away. The bright blue objects in the image are stars located between Earth and the nebula.

Stellar recycling

Materials shed from dying stars join the interstellar medium (the name for gas and dust that exists in the space between stars). From there, these materials are recycled into making new stars. Soon after the Big Bang, the Universe contained only the lightest chemical elements: mostly hydrogen and helium. Nearly all other, heavier, elements—such as carbon and oxygen—have been made since then, in stars or in supernova explosions. Through the formation, evolution, and deaths of stars, these heavier elements have gradually become more abundant in the cosmos. Astronomers call the degree to which a star is rich in heavy elements its "metallicity." Young stars tend to have the highest metallicities, as they contain materials that have already been recycled through several star generations.



▷ **Longer-term cycle**
Stars form partly from materials shed by previous generations of stars. Furthermore, the deaths of massive stars in supernova explosions can trigger changes within the interstellar medium—particularly within star-forming nebulae—that lead to the formation of new stars.